Abdelrahman Sayed Elbeltagy

Instrumentation & Field Engineer

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https://linkedin.com/in/abdelrahman-sayed-93430b1b5 | Military Service: Completed (2023-2025)

PROFESSIONAL SUMMARY

Instrumentation & Field Engineer with hands-on experience in installation, commissioning, and troubleshooting of industrial measurement systems including flowmeters, pressure transmitters, and level sensors. Skilled in 4-20 mA loop calibration, MODBUS protocol, PLC interfacing, and on-site project supervision. Experienced with GSM remote monitoring and LoRaWAN-based Smart Metering applications. Strong background in field service engineering, technical reporting, and customer communication.

PROFESSIONAL EXPERIENCE

Instrumentation & Field Engineer

MAC Company - Instrumentation & Control Systems | Abu Rawash, Giza | Nov 2025 - Mar 2026

- Installed, configured, and commissioned flowmeters, level transmitters, and pressure instruments across 5+ industrial sites.
- Performed 4-20 mA signal calibration and loop checking, maintaining measurement accuracy across all commissioned systems.
- Implemented MODBUS protocol for device communication and data exchange within industrial control systems.
- Integrated GSM modules and LoRaWAN systems for remote monitoring and Smart Metering field applications.
- Diagnosed and resolved instrumentation faults on-site, reducing repeat failure rates by approximately 30%.
- Supervised field installation activities at 4+ sites in the 6th of October area, ensuring full compliance with technical specifications.
- Performed PLC device interfacing and troubleshooting in coordination with control systems engineering teams.
- Delivered after-sales technical support, managing customer communication and resolving reported field issues.
- Produced technical reports, service documentation, and installation handover packages upon project completion.

Mandatory Military Service

Egyptian Armed Forces | Jun 2023 - Oct 2025

- Completed compulsory military service following graduation. Developed discipline, leadership, and team coordination skills.

Mechanical Designer

Pyramakers | Jan 2022 - Mar 2022

- Designed and assembled mechanical components using SolidWorks for manufacturing applications.
- Supported equipment maintenance and upgrades, achieving a 15% reduction in part production time.

PROJECTS

Autonomous Vacuum Robot - Graduation Project

MSA University | 2023

- Designed and built an autonomous robot using SLAM, path planning, and obstacle avoidance algorithms.
- Implemented navigation stack on ROS (Linux) running on Raspberry Pi, integrating LiDAR and ultrasonic sensors for real-time mapping and obstacle detection.
- Tuned sensor fusion pipeline to improve localization accuracy and ensure reliable path execution in dynamic environments.

Remotely Operated Vehicle (ROV) - Competition

MSA University | Nov 2021 - Jul 2022

- Developed an underwater ROV integrating propulsion, sensor systems, and stability control using Arduino.
- Built control and communication system on Raspberry Pi, handling real-time data processing and vehicle maneuvering commands.
- Designed thruster configuration and tested stability under varying load conditions to meet competition performance requirements.

TECHNICAL SKILLS

Instrumentation: Flowmeters | Level Transmitters | Pressure Transmitters | 4-20 mA Calibration | Loop Checking | Smart Metering | Field Service

Protocols & Comms: MODBUS | LoRaWAN | GSM Modules | Remote Monitoring | Long-Range Communication

Control Systems: PLC Interfacing | PLC Troubleshooting | SCADA (basic) | Control Panel Wiring

CAD & Simulation: SolidWorks | AutoCAD | MATLAB

Programming: Python | C++ | ROS | Arduino

Tools: Microsoft Excel (Advanced) | Technical Report Writing | Site Supervision | Customer Communication

EDUCATION

Bachelor of Engineering - Mechatronics
MSA University, Egypt | Graduated: 2023

COURSES & CERTIFICATIONS

- Introduction to Python Programming - Harvard CS50 (2020).

LANGUAGES

Arabic: Native | English: Upper-Intermediate (B2)